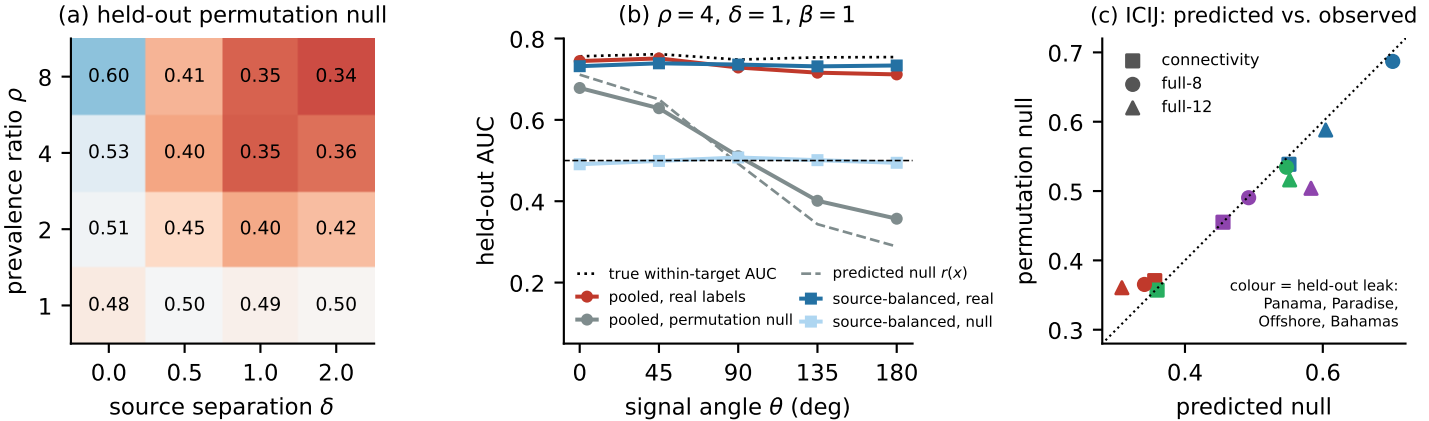
(a) held-out permutation null
prevalence ratio ρ
8
4
2
1
0.60
0.41
0.35
0.34
0.53
0.40
0.35
0.36
0.51
0.45
0.40
0.42
0.48
0.50
0.49
0.50
0.0
0.5
1.0
2.0
source separation δ
(b) ρ = 4, δ = 1, β = 1
held-out AUC
0.8
0.6
0.4
0.2
true within-target AUC
pooled, real labels
pooled, permutation null
predicted null r(x)
source-balanced, real
source-balanced, null
0
45
90
135
180
signal angle θ (deg)
(c) ICIJ: predicted vs. observed
permutation null
0.7
0.6
0.5
0.4
0.3
connectivity
full-8
full-12
colour = held-out leak:
Panama, Paradise,
Offshore, Bahamas
0.4
0.6
predicted null